

## Lecture 5

# The Liouville–Arnold theorem

**Objectives.** Course map: G1, step 5 of 9. *The central theorem of Part I. On a  $2n$ -dimensional phase space, take  $n$  integrals in involution. On a compact connected component of one of their common level sets, where they are independent at every point, the motion lies on an  $n$ -dimensional torus and is a straight line on it, traversed at constant speed. We state the theorem with its hypotheses visible, prove it in six stages, and say at which stage each hypothesis is spent. Without compactness, but with complete flows, the same stages give a cylinder  $T^k \times \mathbb{R}^{n-k}$  instead, which is the case the Toda ring of [Lecture 9](#) needs.*

## 5.1 The question

Functions in involution have commuting Hamiltonian vector fields, whose flows commute where they are complete ([Corollary 2.6](#)), and at most  $n$  independent functions can be in involution ([Proposition 4.11](#)). A system with  $n$  of them has as many conserved quantities as it can have. So the question is

*what do  $n$  independent integrals in involution force the motion to be?*

The answer, [Theorem 5.7](#), concerns a connected component of a common level set on which the integrals are independent at every point. If that component is compact, the motion is a straight line on a torus; if it is not, but the flows on it are complete, it is a straight line on a cylinder. The worked example is the two-dimensional oscillator ([Example 5.3](#)), whose level sets with both energies positive are the tori of [Example 4.15](#).

## 5.2 Independence and involution

A *first integral* of  $H$  is a function  $F$  with  $\{F, H\} = 0$  ([Definition 2.3](#)). Each one confines the motion to a level set of  $F$  — near any point where  $dF \neq 0$ , a hypersurface of dimension  $2n - 1$ . With several first integrals  $F_1, \dots, F_k$  we collect them into one map,

$$F = (F_1, \dots, F_k): \mathcal{M} \rightarrow \mathbb{R}^k, \quad \mathcal{M}_f = F^{-1}(f) = \{x : F_i(x) = f_i \text{ for all } i\}, \quad (5.1)$$

and the motion stays in the level set  $\mathcal{M}_f$  through its initial point. Two things can go wrong with such a collection, and each has a name.

**Definition 5.1 (Independence).**  $F_1, \dots, F_k$  are *independent at  $x$*  if  $dF_1, \dots, dF_k$  are linearly independent at  $x$  — equivalently, if the  $k \times 2n$  Jacobian matrix of  $F$  has rank  $k$  there. They are *independent* if this holds on a dense open subset of  $\mathcal{M}$ .

Independence at a point is what the implicit function theorem needs: if it holds at  $x \in \mathcal{M}_f$ , then near  $x$  the level set  $\mathcal{M}_f$  is a smooth submanifold of dimension  $2n - k$ , with tangent space

$$T_x \mathcal{M}_f = \{v : dF_i(v) = 0 \text{ for all } i\}. \quad (5.2)$$

Asking for it only on a dense open set is deliberate: in every example it fails somewhere, and the places where it fails are the interesting ones.

**Definition 5.2 (Involution).**  $F_1, \dots, F_k$  are in involution if  $\{F_i, F_j\} = 0$  for all  $i, j$ .

**Example 5.3 (The two-dimensional oscillator).** On  $\mathbb{R}^4$  take  $H = H_1 + H_2$  with  $H_i = \frac{1}{2}(p_i^2 + \nu_i^2 q_i^2)$ ,  $\nu_i > 0$ . The  $H_i$  are in involution, since they involve disjoint variables. Their Jacobian is

$$\begin{pmatrix} \nu_1^2 q_1 & 0 & p_1 & 0 \\ 0 & \nu_2^2 q_2 & 0 & p_2 \end{pmatrix} \quad (\text{columns } q_1, q_2, p_1, p_2),$$

of rank 2 unless one row vanishes, that is unless  $q_1 = p_1 = 0$  or  $q_2 = p_2 = 0$ . Those are two planes in  $\mathbb{R}^4$ , closed with empty interior, so the  $H_i$  are independent. The level sets are pictured in Figure 5.1: the image of  $F = (H_1, H_2)$  is the closed quadrant, over each interior point sits a torus, and over the edges — exactly where independence fails — the torus collapses to a circle or a point.

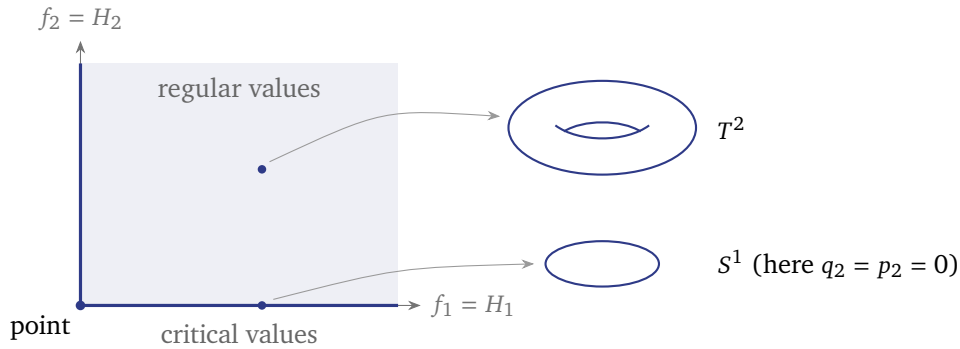


Figure 5.1: The level sets of  $F = (H_1, H_2)$  for the two-dimensional oscillator: a torus over each interior point of the quadrant, a circle over each edge point, where the rank of  $dF$  drops, and a point over the corner. This is the picture the Liouville–Arnold theorem generalises.

Each of the two conditions is easy to have without the other.

**Example 5.4 (Independent but not in involution).** For a particle in a central potential on  $\mathbb{R}^3$ ,  $H = \frac{1}{2}|p|^2 + V(|q|)$ , the three components of  $\ell = q \times p$  are first integrals. So  $H, \ell_1, \ell_2$  are three first integrals on a 6-dimensional phase space, and they are independent. They are not in involution:

$$\{\ell_1, \ell_2\} = \ell_3, \quad (5.3)$$

and cyclically. Proposition 4.11 says three commuting independent functions is the most  $n = 3$  allows; three non-commuting ones are not the same thing.

### 5.3 Liouville integrable systems: the algebraic definition

**Definition 5.5 (Liouville integrable).** A Hamiltonian system  $(M, \omega, H)$  with  $\dim M = 2n$  is *Liouville integrable* if there are  $n$  independent functions  $F_1, \dots, F_n$  in involution such that  $H$  is a function of them,  $H = h(F_1, \dots, F_n)$ . Most often  $H$  is one of them,  $H = F_1$ .

By Proposition 4.11,  $n$  is the most one can ask for.

Two remarks on the definition. First,  $H = h(F)$  is automatically a first integral of each  $F_i$ , since it is built from functions that commute with  $F_i$ ; and by the chain rule  $dH = \sum_i \partial_i h dF_i$ , so

$$X_{h(F)} = \sum_i \frac{\partial h}{\partial f_i}(F) X_{F_i}. \quad (5.4)$$

Second, the functions are part of the data but not canonical:  $(H_1, H_2)$  and  $(H, H_2)$  in Example 5.3 define the same level sets.

**Example 5.6 (Integrable systems we already own).** 1.  $n = 1$ . Every  $H$  with  $dH \neq 0$  on a dense open set.

2. The  $n$ -dimensional oscillator, with  $F_i = H_i$  as in Example 5.3.

3. A central force in  $\mathbb{R}^3$ , with  $F = (H, |\ell|^2, \ell_3)$ . Involution:  $\{H, \ell_i\} = 0$  for every  $i$ , and  $\{|\ell|^2, \ell_3\} = 0$  by (5.3). Independence: the  $p$ -components of  $d\ell_i$  are  $e_i \times q$ , all orthogonal to  $q$ , while that of  $dH$  is  $p$ ; so  $dH$  is independent of the other two wherever  $q \cdot p \neq 0$ , and  $d|\ell|^2 = 2 \sum_i \ell_i d\ell_i$  is independent of  $d\ell_3$  wherever  $\ell$  is not parallel to  $e_3$ . Both conditions fail only on the zero set of a nonzero polynomial.

### 5.4 The theorem, hypothesis by hypothesis

**Theorem 5.7 (Liouville–Arnold · proved modulo Lemma 5.10 and Theorem 1.9).** Let  $(M, \omega)$  be a symplectic manifold of dimension  $2n$ , and  $F_1, \dots, F_n$  functions in involution. Fix  $f \in \mathbb{R}^n$ , let  $\Sigma$  be a connected component of the level set  $M_f$ , and suppose

(I)  $dF_1, \dots, dF_n$  are linearly independent at every point of  $\Sigma$ ; and

(C)  $\Sigma$  is compact.

Then:

1.  $\Sigma$  is a smooth Lagrangian submanifold, invariant under the flow of every  $X_{F_i}$ ;
2.  $\Sigma$  is diffeomorphic to the torus  $T^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n$ , with angle coordinates  $\varphi_1, \dots, \varphi_n$  defined modulo  $2\pi$ ;
3. for any  $H = h(F)$ , the motion on  $\Sigma$  is  $\varphi(t) = \varphi(0) + \nu t$ , with a frequency vector  $\nu \in \mathbb{R}^n$  that depends on  $f$  only.

Deleting any one hypothesis gives a counterexample; three are exercises. The one to see now is compactness, the one the course will not have. The free particle  $H = \frac{1}{2}p^2$  on  $\mathbb{R}^2$  is integrable, with  $n = 1$ . At  $E > 0$  each component of its level set is a line  $p = \pm\sqrt{2E}$ , and the motion on it is not periodic: this is  $T^0 \times \mathbb{R}^1$ . When compactness fails but the flows on  $\Sigma$  are complete, stage (iv) of the proof shows that a cylinder of this kind is the shape of the conclusion; without completeness there is no such classification. The Toda ring of Lecture 9 is the case the course needs it for.

## 5.5 The proof, in six stages

The proof turns  $n$  commuting flows into an action of the group  $\mathbb{R}^n$  on  $\Sigma$ , and then identifies  $\Sigma$  with a quotient of  $\mathbb{R}^n$ . Groups acting on spaces have not appeared in the course so far, so stage (0) sets up the three words the proof uses.

### Stage (0): an action of $\mathbb{R}^n$ , its orbits and its stabiliser

**Definition 5.8 (Action, orbit, stabiliser).** An *action* of  $\mathbb{R}^n$  on a set  $\Sigma$  is a family of maps  $\Phi_t : \Sigma \rightarrow \Sigma$ ,  $t \in \mathbb{R}^n$ , with

$$\Phi_0 = \text{id}, \quad \Phi_{s+t} = \Phi_s \circ \Phi_t.$$

The *orbit* of  $x$  is  $\{\Phi_t(x) : t \in \mathbb{R}^n\}$ ; the action is *transitive* if some (hence every) orbit is all of  $\Sigma$ . The *stabiliser* of  $x$  is  $\Lambda_x = \{t : \Phi_t(x) = x\}$ .

**Lemma 5.9.**  $\Lambda_x$  is a subgroup of  $\mathbb{R}^n$ , the same for every point of the orbit of  $x$ , and  $t \bmod \Lambda_x \mapsto \Phi_t(x)$  is a bijection from  $\mathbb{R}^n / \Lambda_x$  onto the orbit.

*Proof.* If  $s, t \in \Lambda_x$  then  $\Phi_{s+t}(x) = \Phi_s(\Phi_t(x)) = x$ , and  $\Phi_{-t}(x) = \Phi_{-t}(\Phi_t(x)) = x$ . If  $y = \Phi_s(x)$  and  $t \in \Lambda_x$ , then  $\Phi_t(y) = \Phi_{t+s}(x) = \Phi_s(\Phi_t(x)) = y$  — here we used  $s + t = t + s$ : the group is abelian. Finally  $\Phi_s(x) = \Phi_t(x)$  if and only if  $\Phi_{s-t}(x) = x$ , i.e.  $s - t \in \Lambda_x$ . ■

Two examples fix the words (Figure 5.2). The oscillator flow is an action of  $\mathbb{R}$  on the plane; the orbit of a point other than the origin is a circle with stabiliser  $(2\pi/\nu)\mathbb{Z}$ , while the origin is its own orbit, with stabiliser all of  $\mathbb{R}$ . And on the torus  $\mathbb{R}^2/2\pi\mathbb{Z}^2$  the flow along the direction  $(1, \sqrt{2})$  is an action of  $\mathbb{R}$  with stabiliser  $\{0\}$ : the orbit is a copy of  $\mathbb{R}$  that never closes, winds densely (Lecture 6), and is still not the whole torus. One flow cannot fill a two-dimensional space; the proof uses  $n$  flows on an  $n$ -dimensional one.

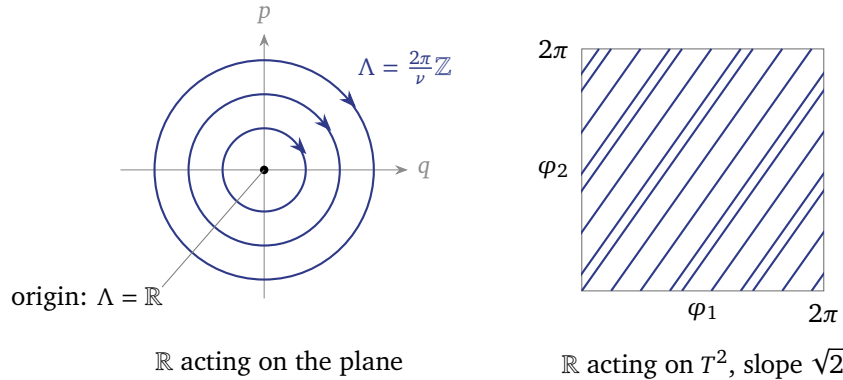


Figure 5.2: Orbits and stabilisers: the two examples of the text. Left, the oscillator flow on the plane; right, the flow of slope  $\sqrt{2}$  on the torus (a square with opposite sides identified), over ten crossings of the bottom edge.

### Stage (i): $\Sigma$ is a Lagrangian submanifold

By (I) and the implicit function theorem, near each point of  $\Sigma$  the level set is a smooth connected piece of dimension  $2n - n = n$ , which therefore lies in  $\Sigma$ ; so  $\Sigma$  is a smooth

$n$ -dimensional submanifold with tangent spaces (5.2). The fields  $X_{F_j}$  are tangent to it, because

$$dF_i(X_{F_j}) = X_{F_j}F_i = \{F_i, F_j\} = 0. \quad (5.5)$$

At each point they are linearly independent, since  $\iota_{X_{F_j}}\omega = dF_j$  and  $v \mapsto \iota_v\omega$  is injective (3.10). So the  $n$  vectors  $X_{F_1}, \dots, X_{F_n}$  are a basis of the  $n$ -dimensional tangent space of  $\Sigma$  at every point (Figure 5.3). On that basis  $\omega(X_{F_i}, X_{F_j}) = \{F_i, F_j\} = 0$ , so  $\omega$  vanishes on  $T\Sigma$ :  $\Sigma$  is Lagrangian (Definition 4.10). *Spent here: independence, and involution twice.*

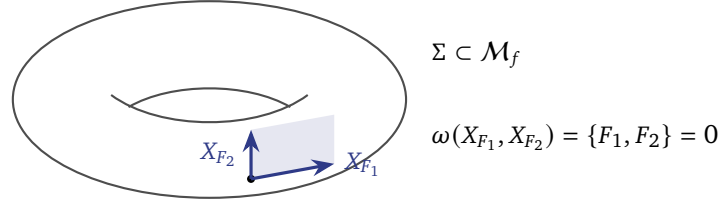


Figure 5.3: A regular compact level set with  $n = 2$ . The Hamiltonian vector fields of the integrals are tangent to it (5.5), independent, and therefore span each tangent plane;  $\omega$  vanishes on the parallelogram they span. Stage (i) in one picture.

### Stage (ii): $n$ complete commuting flows, hence an action of $\mathbb{R}^n$

The fields  $X_{F_i}$  are tangent to  $\Sigma$ , so they restrict to vector fields on the compact manifold  $\Sigma$ , and by Proposition 1.7 they are complete there: each flow  $\varphi_s^i$  is defined on  $\Sigma$  for all  $s \in \mathbb{R}$  and maps  $\Sigma$  to itself. By Corollary 2.6 these flows commute. Define

$$\Phi_t = \varphi_{t_1}^1 \circ \varphi_{t_2}^2 \circ \dots \circ \varphi_{t_n}^n, \quad t \in \mathbb{R}^n. \quad (5.6)$$

Then  $\Phi_0 = \text{id}$ , and  $\Phi_{s+t} = \Phi_s \circ \Phi_t$  because the factors commute and each is a one-parameter group. This is an action of  $\mathbb{R}^n$  on  $\Sigma$ . *Spent here: compactness, for completeness, and involution, for commutativity.*

### Stage (iii): the action is transitive

Fix  $x_0 \in \Sigma$  and consider the orbit map  $\Phi^{x_0}: \mathbb{R}^n \rightarrow \Sigma$ ,  $t \mapsto \Phi_t(x_0)$ . Since the factors in (5.6) commute, any one of them may be put first, so

$$\frac{\partial}{\partial t_i} \Phi_t(x_0) = X_{F_i}(\Phi_t(x_0)).$$

The derivative of  $\Phi^{x_0}$  at every  $t$  carries the standard basis of  $\mathbb{R}^n$  to a basis of the tangent space (stage (i)), so by the inverse function theorem  $\Phi^{x_0}$  is a local diffeomorphism: it maps a small ball about any  $t$  diffeomorphically onto a neighbourhood of  $\Phi_t(x_0)$  (Figure 5.4).

Consequently every orbit is open in  $\Sigma$ . The orbits partition  $\Sigma$ , so the complement of one orbit is a union of others and is open too; an orbit is therefore open and closed.  $\Sigma$  is connected, so there is only one orbit. *Spent here: independence (the local diffeomorphism), connectedness, and involution once more (the factors commute).*

By Lemma 5.9, with  $\Lambda = \Lambda_{x_0}$  now the same for every point of  $\Sigma$ ,

$$\mathbb{R}^n / \Lambda \longrightarrow \Sigma, \quad t \bmod \Lambda \longmapsto \Phi_t(x_0) \text{ is a bijection.} \quad (5.7)$$

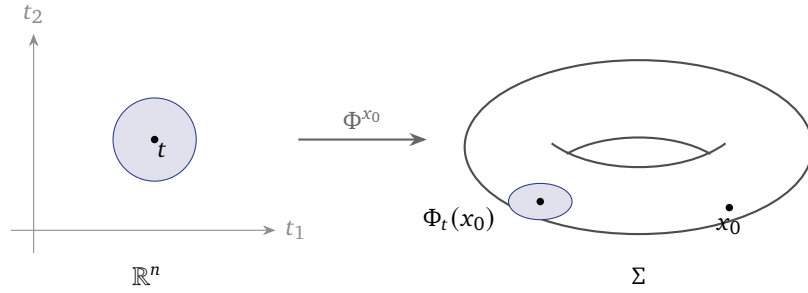


Figure 5.4: Stage (iii). Because the  $X_{F_i}$  are a basis everywhere, the orbit map carries a small ball about every  $t$  diffeomorphically onto a patch of  $\Sigma$ . So each orbit is open; in a connected  $\Sigma$  an open orbit whose complement is also open is everything. The map is not injective — that is stage (iv).

### Stage (iv): $\Lambda$ is a lattice of full rank

This is the step doing the most work, and it has two halves that must not be merged.

$\Lambda$  is discrete. Choose a ball  $U$  about 0 on which  $\Phi^{x_0}$  is injective (stage (iii)). If  $\lambda \in \Lambda \cap U$  then  $\Phi^{x_0}(\lambda) = x_0 = \Phi^{x_0}(0)$ , so  $\lambda = 0$ . Translating, for any  $\lambda \in \Lambda$  the ball  $\lambda + U$  contains no other point of  $\Lambda$ , because  $\Lambda - \lambda = \Lambda$ . Every point of  $\Lambda$  is isolated. Contrast  $\mathbb{Z} + \sqrt{2}\mathbb{Z} \subset \mathbb{R}$ , a subgroup with points arbitrarily close to 0: what excludes that here is independence, through the local diffeomorphism.

**Lemma 5.10** (Discrete subgroups of  $\mathbb{R}^n$  · cited). A discrete subgroup of  $\mathbb{R}^n$  is  $\mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_k$  for some linearly independent  $e_1, \dots, e_k$ ,  $0 \leq k \leq n$ . (Arnold, §49 D, Lemma 3.)

Discreteness alone gives a cylinder, not a torus. Complete  $e_1, \dots, e_k$  to a basis  $e_1, \dots, e_n$  of  $\mathbb{R}^n$ . Then

$$\mathbb{R}^k / \mathbb{Z}^k \times \mathbb{R}^{n-k} \longrightarrow \Sigma, \quad (\theta, y) \longmapsto \Phi^{x_0} \left( \sum_{i \leq k} \theta_i e_i + \sum_{j > k} y_j e_j \right)$$

is well defined and bijective by (5.7), and a local diffeomorphism by stage (iii). A bijective local diffeomorphism is a diffeomorphism. So  $\Sigma \cong T^k \times \mathbb{R}^{n-k}$ , and every  $k$  from 0 to  $n$  is possible for a discrete stabiliser.

Compactness forces  $k = n$ .  $T^k \times \mathbb{R}^{n-k}$  is compact only if  $k = n$ , and  $\Sigma$  is compact. Hence  $\Lambda$  has full rank and  $\Sigma \cong \mathbb{R}^n / \Lambda \cong T^n$ . Spent here: independence for discreteness, compactness for full rank — the second use of compactness, and the one that fails for the Toda ring of Lecture 9.

With  $n = 1$  (Figure 5.5) the two cases are a circle,  $\Lambda = \tau\mathbb{Z}$  with  $\tau$  the period, and a line,  $\Lambda = \{0\}$ , which is the free particle’s level set.

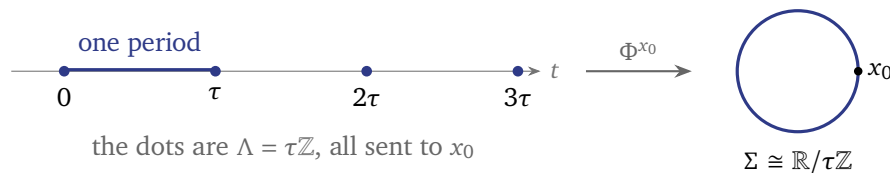


Figure 5.5: Stage (iv) with  $n = 1$ . The orbit map sends every point of the stabiliser  $\Lambda = \tau\mathbb{Z}$  to  $x_0$  and wraps each interval of length  $\tau$  once round  $\Sigma$ . Discreteness allows  $\Lambda = \{0\}$  as well, and then the line is not wrapped at all; only compactness rules that out.

### Stage (v): the period lattice and a basis of cycles

**Definition 5.11 (Period lattice).** The stabiliser  $\Lambda \subset \mathbb{R}^n$  of stage (iv) is the *period lattice* of  $\Sigma$ . A basis  $e_1, \dots, e_n$  of it gives *basis cycles*  $\gamma_i(s) = \Phi_{se_i}(x_0)$ ,  $s \in [0, 1]$ , closed loops on  $\Sigma$ .

A point  $t \in \mathbb{R}^n$  comes back to  $x_0$  exactly when  $t \in \Lambda$ , so the components of  $t$  in the basis  $e_i$ , times  $2\pi$ , are angles. Write  $A = (e_1 \cdots e_n)$  for the matrix whose columns are the basis vectors, and set

$$\varphi = 2\pi A^{-1}t \pmod{2\pi\mathbb{Z}^n}. \quad (5.8)$$

These are the angle coordinates of [Theorem 5.7](#):  $\varphi \mapsto \Phi_{A\varphi/2\pi}(x_0)$  identifies  $\mathbb{R}^n/2\pi\mathbb{Z}^n$  with  $\Sigma$  ([Figure 5.6](#)), and along  $\gamma_j$  the angle  $\varphi_j$  runs once from 0 to  $2\pi$  while the others are constant.

Here is [Lecture 3](#)'s annulus again. Each  $\varphi_i$  is not a function on  $\Sigma$ , only a function modulo  $2\pi$ , but  $d\varphi_i$  is a well-defined 1-form, closed since it is locally  $d$  of a function. It is *not exact*:

$$\oint_{\gamma_j} d\varphi_i = 2\pi \delta_{ij} \neq 0 \quad (i = j),$$

whereas an exact form integrates to zero round every loop ([Proposition 3.10](#)). The torus carries  $n$  independent closed non-exact 1-forms, detected by  $n$  loops; [Lecture 6](#) integrates  $p \cdot dq$  over those same loops.

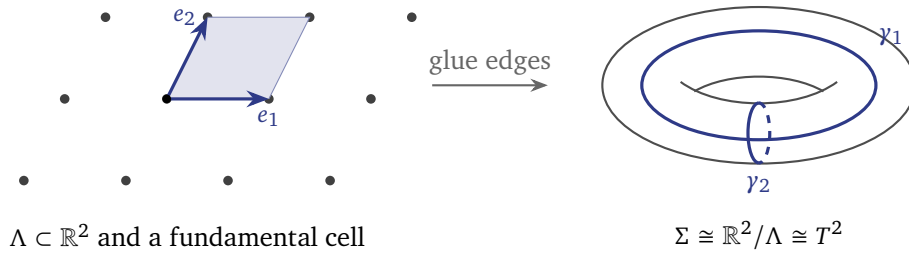


Figure 5.6: Stages (iv)–(v). Left: the period lattice  $\Lambda$  with a basis  $e_1, e_2$  and the cell they span. Every point of  $\mathbb{R}^2$  is a lattice translate of exactly one point of the cell, and gluing its opposite edges gives the torus. Right: the images of the edges,  $t = se_1$  and  $t = se_2$  for  $s \in [0, 1]$ , are the basis cycles  $\gamma_1, \gamma_2$  — the cycles of [Figure 3.6](#).

### Stage (vi): straight-line motion, and what is left for [Lecture 6](#)

Let  $H = h(F)$ . On  $\Sigma$  the coefficients in (5.4) are the constants  $c_i = \partial_i h(f)$ , so  $X_H = \sum_i c_i X_{F_i}$ , whose flow is  $\Phi_{\tau c}$ : in  $\mathbb{R}^n$  the motion is  $t \mapsto t + \tau c$ . By (5.8),

$$\varphi(\tau) = \varphi(0) + \nu \tau, \quad \nu = 2\pi A^{-1} \nabla h(f). \quad (5.9)$$

That is conclusion 3 of [Theorem 5.7](#), and it completes the proof.

One more fact is needed next time, and it is a consequence of stage (i).

**Proposition 5.12 ( $p \cdot dq$  is closed on the torus).** If  $\mathcal{M}$  is  $\mathbb{R}^{2n}$  or a cotangent bundle, the restriction of  $p \cdot dq$  to  $\Sigma$  is a closed 1-form.

*Proof.* Restriction is pullback by the inclusion, and  $d$  commutes with pullback (3.6). So  $d(p \cdot dq|_{\Sigma}) = (d(p \cdot dq))|_{\Sigma} = -\omega|_{\Sigma} = 0$ , by [Theorem 4.17](#) and stage (i). ■

So  $\oint_{\gamma} p \cdot dq$  over a loop on  $\Sigma$  is unchanged when the loop is deformed within  $\Sigma$  (Corollary 3.13). That is what Lecture 6 needs to define the *action variables*  $I_i = \frac{1}{2\pi} \oint_{\gamma_i} p \cdot dq$  and to show that, as  $f$  varies, the angles (5.8) and the  $I_i$  together form canonical coordinates on a whole neighbourhood of  $\Sigma$ . The theorem as proved here is about one torus at a time; how the tori fit together as  $f$  moves is Lecture 6’s business near one torus, and globally it can be subtle (optional lecture A).

Tutorial

## One torus, two lattices

**Example 5.13 (One torus, two lattices).** Return to Example 5.3. The flow of  $X_{H_i}$  turns the  $i$ -th  $(q_i, p_i)$ -plane clockwise at angular speed  $\nu_i$  and leaves the other alone.

- With  $F = (H_1, H_2)$ ,  $\Phi_t$  turns the planes through  $\nu_1 t_1$  and  $\nu_2 t_2$ , so  $\Lambda = \frac{2\pi}{\nu_1} \mathbb{Z} \oplus \frac{2\pi}{\nu_2} \mathbb{Z}$ , a rectangular lattice.
- With  $F = (H, H_2)$ ,  $X_H$  turns both planes and  $X_{H_2}$  the second, so  $\Phi_t$  turns them through  $\nu_1 t_1$  and  $\nu_2(t_1 + t_2)$ . Now  $t \in \Lambda$  when  $\nu_1 t_1$  and  $\nu_2(t_1 + t_2)$  are both in  $2\pi\mathbb{Z}$ , and a basis is

$$e_1 = \frac{2\pi}{\nu_1}(1, -1), \quad e_2 = \frac{2\pi}{\nu_2}(0, 1).$$

Unless  $\nu_2/\nu_1$  is an integer,  $(2\pi/\nu_1, 0) \notin \Lambda$ : the lattice is not the product lattice of the first bullet (Figure 5.7).

The level set is the same torus in both cases, and the two lattices have the same cell area  $4\pi^2/(\nu_1\nu_2)$ ; they describe the same torus with different generators of the action. The basis cycles are the same too: in both cases  $\gamma_1$  goes once round the first plane and  $\gamma_2$  once round the second.

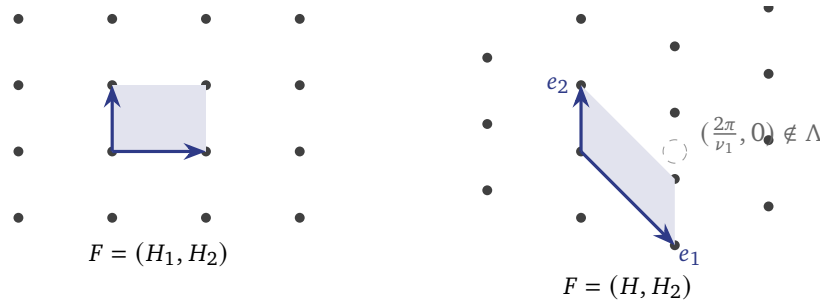


Figure 5.7: The period lattice of one oscillator torus,  $\nu_1 = 1$ ,  $\nu_2 = \sqrt{2}$ , for two choices of integrals. Left: rectangular. Right: the lattice of Example 5.13, a different lattice with the same cell area. The lattice records the generators as well as the torus; the frequency vector read off it in stage (vi) does not depend on the choice.

By (5.9), with  $H = H_1 + H_2$ : the product lattice and  $\nabla h = (1, 1)$  give  $\nu = (\nu_1, \nu_2)$ ; the second lattice and  $H = F_1$ ,  $\nabla h = (1, 0)$ , give  $\nu = (\nu_1, \nu_2)$  again. The lattice depends on the choice of integrals; the motion does not.

## 5.6 What it buys

- *Tori, or cylinders.* If the flows are complete on  $\Sigma$  — which compactness guarantees but need not be what supplies it — stages (i)–(iv) give  $\Sigma \cong T^k \times \mathbb{R}^{n-k}$ , and compactness is needed only to force  $k = n$ . For the Toda ring, whose level sets are not compact, [Lecture 9](#) proves completeness separately and finds  $T^{N-1} \times \mathbb{R}$ .
- *Angles and cycles.* The period lattice gives angle coordinates,  $n$  basis cycles and a frequency vector (5.9); and, in  $\mathbb{R}^{2n}$  or a cotangent bundle,  $p \cdot dq$  is closed on the torus ([Proposition 5.12](#)). [Lecture 6](#) integrates it over the cycles to get the actions, and completes the angles to canonical coordinates.

## 5.7 Caveats, in one place

- *Status.* [Theorem 5.7](#) is proved modulo the cited [Lemma 5.10](#) and [Theorem 1.9](#). Stage (ii) also uses [Corollary 2.6](#) on  $\mathcal{M}$ . On  $\mathbb{R}^{2n}$  and on cotangent bundles, where every cotangent chart is canonical ([Theorem 4.17](#)), the coordinate proof of [Lecture 2](#) gives it. On a general symplectic manifold it rests on the Jacobi identity for the bracket of  $\omega$  and the second proof of the engine, both proved in readings ([Lecture 4](#), [Lecture 2](#)).
- *Independent on a dense set, or on all of  $\Sigma$ .* The definition allows the integrals to be dependent somewhere; the oscillator's are, on two planes. The theorem applies only to the components of level sets on which they are independent at every point.
- *One torus at a time.* The theorem is about one level set. How the tori fit together as  $f$  varies is [Lecture 6](#)'s business near one torus; globally it can be subtle (optional lecture A).

### Summary.

- *The definition* is algebraic:  $n$  independent functions in involution on a  $2n$ -dimensional phase space, with  $H$  a function of them ([Definition 5.5](#)).
- *The theorem* ([Theorem 5.7](#)): a compact connected component of a level set, on which the differentials are independent, is a Lagrangian torus  $T^n$ , and the motion on it is  $\varphi(t) = \varphi(0) + \nu t$ .
- *The proof:* tangent commuting flows, complete by compactness, give an action of  $\mathbb{R}^n$ ; independence and connectedness make it transitive, with a discrete stabiliser  $\Lambda$ ; the quotient is  $T^k \times \mathbb{R}^{n-k}$ , and compactness forces  $k = n$ .
- *The period lattice*  $\Lambda$  gives angles, basis cycles and frequencies (5.9); in  $\mathbb{R}^{2n}$  or a cotangent bundle,  $p \cdot dq$  is closed on the torus ([Proposition 5.12](#)).
- *Next:* [Lecture 6](#) integrates  $p \cdot dq$  over the basis cycles, and finds canonical action–angle coordinates near the torus.

**Tutorial.** Exercise Sheet 5.

## References

Arnold, *Mathematical Methods of Classical Mechanics*, §49 (the theorem and its proof; Lemma 2 of §49 C is our stage (iii), Lemma 3 of §49 D our [Lemma 5.10](#)); Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 2 (the theorem, and the examples).

**Conventions.** Arnold writes  $\omega = \sum dp \wedge dq$ , the negative of ours; nothing in this lecture depends on the sign of  $\omega$ , since every use of it is a statement that something vanishes.